

Chapter 6: Orthogonal Polynomials

Mathematical Physics

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Outline

- 1 From Laplace to ordinary Legendre
- 2 Legendre polynomials
- 3 Associated Legendre and spherical harmonics
- 4 Hermite polynomials
- 5 Laguerre polynomials
- 6 Chebyshev polynomials
- 7 Summary

Where we are going

This chapter studies four families of **orthogonal polynomials** that arise by separating variables in Laplace, Poisson, Helmholtz, and Schrödinger problems. “Orthogonal” means a weighted inner product of distinct-degree polynomials vanishes; the interval and weight depend on the family.

- **Legendre** P_n . From Laplace in spherical coordinates to the ordinary ($m = 0$) Legendre equation, series termination, generating function, recurrence, Rodrigues, orthogonality, and Fourier–Legendre expansions. Associated Legendre functions and spherical harmonics Y_n^m for $m \neq 0$, with the degree n the same integer that the axisymmetric problem picks out through $\lambda = n(n + 1)$.
- **Hermite** H_n . The Hermite equation and series solution, quantum harmonic oscillator motivation, generating function, Rodrigues formula, and weighted orthogonality.
- **Laguerre** L_n and **Chebyshev** T_n . Defining ODEs, first polynomials, weights and orthogonality, recurrences, and useful formula sheets.

A comparison table at the end collects interval, weight, and ODE for each family.

Table of contents

- 1 From Laplace to ordinary Legendre
- 2 Legendre polynomials
- 3 Associated Legendre and spherical harmonics
- 4 Hermite polynomials
- 5 Laguerre polynomials
- 6 Chebyshev polynomials
- 7 Summary

Two-dimensional Laplace equation in polar coordinates

Step 1: separate the angular and radial parts. A function satisfying Laplace's equation $\nabla^2\psi = 0$ is called a *harmonic function*. The operator ∇^2 is the Laplacian: $\psi_{xx} + \psi_{yy}$ in Cartesian coordinates. In polar coordinates r is radius and θ is angle. We use its coordinate form without deriving it:

$$\nabla^2\psi = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial\psi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2\psi}{\partial\theta^2} = 0. \quad (6.1.1)$$

Take a separated solution $\psi(r, \theta) = u(r)v(\theta)$. Then

$$\frac{\partial\psi}{\partial r} = u'(r)v(\theta), \quad \frac{\partial^2\psi}{\partial\theta^2} = u(r)v''(\theta),$$

so substitution into (6.1.1) gives

$$\frac{v}{r} \frac{d}{dr} (ru') + \frac{u}{r^2} v'' = 0.$$

Multiplying by $r^2/(uv)$ yields

$$\frac{r(ru')'}{u} + \frac{v''}{v} = 0,$$

Two-dimensional Laplace equation in polar coordinates (cont.)

and therefore

$$-\frac{v''}{v} = \frac{r(ru')'}{u} = \lambda, \quad (6.1.2)$$

because the left member depends only on θ and the right member only on r .

The angular equation is

$$v'' + \lambda v = 0.$$

For $\lambda > 0$, write $\lambda = \mu^2$; then

$$v(\theta) = A \cos(\mu\theta) + B \sin(\mu\theta).$$

The periodicity condition $v(\theta + 2\pi) = v(\theta)$ restricts μ : since $\cos \mu\theta$ and $\sin \mu\theta$ have period $2\pi/\mu$, the condition can hold for all θ only when 2π is a whole multiple of that period, i.e. only when μ is an integer. Hence $\mu = n$ with $n \in \mathbb{Z}$. Since the modes n and $-n$ span the same real sine-cosine pair, we may take $n \geq 0$ and $\lambda = n^2$. If $\lambda = 0$, then $v = A + B\theta$, and periodicity forces $B = 0$; this is the $n = 0$ mode. If $\lambda < 0$, the angular solutions are exponentials rather than periodic functions, so no nonzero 2π -periodic mode occurs.

The radial equation is

$$r(ru')' - n^2 u = 0.$$

Two-dimensional Laplace equation in polar coordinates (cont.)

Expanding the derivative gives a Cauchy–Euler (equidimensional) equation — one in which every term carries as many powers of r as derivatives of u —

$$r^2 u'' + ru' - n^2 u = 0, \quad (6.1.3)$$

which is therefore solved by trial powers $u = r^p$. For $n \geq 1$, try $u = r^p$. Since

$$u' = pr^{p-1}, \quad u'' = p(p-1)r^{p-2},$$

we obtain

$$r^2 u'' + ru' - n^2 u = [p(p-1) + p - n^2]r^p = (p^2 - n^2)r^p.$$

Thus $p = \pm n$, and

$$u_n(r) = C_n r^n + D_n r^{-n} \quad (n \geq 1).$$

For $n = 0$, (6.1.3) becomes $(ru')' = 0$. Integrating twice,

$$ru' = C_0, \quad u' = \frac{C_0}{r}, \quad u(r) = D_0 + C_0 \log r.$$

Two-dimensional Laplace equation in polar coordinates (cont.)

On an annulus, all these radial solutions are allowed. Inside a disk we must also examine the center. Inside a disk containing $r = 0$, boundedness removes $\log r$ and all r^{-n} terms:

$$\psi(r, \theta) = A_0 + \sum_{n=1}^{\infty} r^n [A_n \cos(n\theta) + C_n \sin(n\theta)]. \quad (6.1.4)$$

Equivalently, with $z = re^{i\theta}$, $c_0 = A_0$, and $c_n = A_n - iC_n$ for $n \geq 1$,

$$\psi(r, \theta) = \operatorname{Re} \left(\sum_{n=0}^{\infty} c_n z^n \right), \quad z^n = r^n e^{in\theta}.$$

Thus the regular real harmonic functions are real parts of analytic power series; exterior or annular domains additionally allow negative powers and, for the angularly independent $n = 0$ mode, a logarithm.

Three-dimensional Laplace equation in spherical coordinates

In spherical coordinates r is radius, θ is the polar angle measured from the positive vertical axis, and ϕ is the azimuthal angle around that axis. We use the spherical Laplacian without deriving this coordinate formula:

$$\nabla^2 \psi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \psi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \psi}{\partial \phi^2} = 0. \quad (6.1.5)$$

Step 1: separate the radial variable. Set $\psi(r, \theta, \phi) = u(r)y(\theta, \phi)$. Because y is independent of r and u is independent of the angles,

$$\begin{aligned} \frac{\partial}{\partial r} (r^2 \psi_r) &= y(r^2 u')', \\ \frac{\partial}{\partial \theta} (\sin \theta \psi_\theta) &= u(\sin \theta y_\theta)_\theta. \end{aligned}$$

Substitution into (6.1.5) and multiplication by $r^2/(uy)$ give

$$\frac{(r^2 u')'}{u} = - \frac{\frac{1}{\sin \theta} (\sin \theta y_\theta)_\theta + \frac{1}{\sin^2 \theta} y_{\phi\phi}}{y} = \lambda. \quad (6.1.6)$$

Three-dimensional Laplace equation in spherical coordinates (cont.)

The first expression depends only on r and the second only on (θ, ϕ) , so each equals a constant λ .

Step 2: radial Euler equation. Expanding $(r^2 u')' = r^2 u'' + 2ru'$ gives

$$r^2 u'' + 2ru' - \lambda u = 0.$$

Try $u = r^\sigma$. Then $u' = \sigma r^{\sigma-1}$, $u'' = \sigma(\sigma-1)r^{\sigma-2}$, and

$$[\sigma(\sigma+1) - \lambda] r^\sigma = 0.$$

So $\lambda = \sigma(\sigma+1)$. Once the angular problem forces $\lambda = n(n+1)$ with $n = 0, 1, 2, \dots$ (next frames), the two exponents are

$$\sigma = n, \quad \sigma = -n-1,$$

and

$$u_n(r) = A_n r^n + B_n r^{-n-1}. \tag{6.1.7}$$

The r^n branch is regular at the origin; the r^{-n-1} branch decays at infinity.

Three-dimensional Laplace equation in spherical coordinates (cont.)

The remaining angular eigenproblem is

$$\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 y}{\partial \phi^2} + \lambda y = 0.$$

We first treat the **axisymmetric** case (no ϕ -dependence), which yields the ordinary Legendre equation and $\lambda = n(n+1)$. Associated Legendre modes for $m \neq 0$ come later as a short extension.

Axisymmetry: the ordinary Legendre equation

Step 1: assume no azimuthal dependence and change variables. If ψ is independent of ϕ (axisymmetry about the polar axis), the angular equation reduces to

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dv}{d\theta} \right) + \lambda v = 0.$$

Set $x = \cos \theta$, so $dx/d\theta = -\sin \theta$ and $d/d\theta = -\sin \theta d/dx$. Then

$$\begin{aligned} \sin \theta \frac{dv}{d\theta} &= -(1-x^2)v_x, \\ \frac{d}{d\theta} \left(\sin \theta \frac{dv}{d\theta} \right) &= \sin \theta \frac{d}{dx} [(1-x^2)v_x], \end{aligned}$$

and therefore

$$\frac{d}{dx} [(1-x^2)v'(x)] + \lambda v(x) = 0. \quad (6.1.8)$$

Expanding the derivative gives the equivalent form

$$(1-x^2)v'' - 2xv' + \lambda v = 0. \quad (6.1.9)$$

(The poles $\theta = 0, \pi$ correspond to the endpoints $x = \pm 1$.)

Axisymmetry: the ordinary Legendre equation (cont.)

Step 2: power series and coefficient recurrence. Set

$$v(x) = \sum_{j=0}^{\infty} a_j x^j.$$

Then

$$v' = \sum_{j=0}^{\infty} (j+1) a_{j+1} x^j,$$

$$v'' = \sum_{j=0}^{\infty} (j+2)(j+1) a_{j+2} x^j,$$

$$-x^2 v'' = - \sum_{j=0}^{\infty} j(j-1) a_j x^j,$$

$$-2xv' = -2 \sum_{j=0}^{\infty} j a_j x^j.$$

Axisymmetry: the ordinary Legendre equation (cont.)

Substitution into (6.1.9) gives

$$\begin{aligned} 0 &= \sum_{j=0}^{\infty} [(j+2)(j+1)a_{j+2} - j(j-1)a_j - 2ja_j + \lambda a_j] x^j \\ &= \sum_{j=0}^{\infty} [(j+2)(j+1)a_{j+2} + (\lambda - j(j+1))a_j] x^j. \end{aligned}$$

Hence the coefficient recurrence is

$$a_{j+2} = -\frac{\lambda - j(j+1)}{(j+2)(j+1)} a_j. \quad (6.1.10)$$

Even and odd chains are independent.

Step 3: select solutions that remain finite at the poles. The points $x = \pm 1$ correspond to the north and south poles of the sphere. A polynomial is finite at both. The recurrence stops at degree n precisely when $\lambda = n(n+1)$; this is the simplest way to construct the acceptable solutions.

We use the standard endpoint result without its proof: a nonzero solution finite at both endpoints exists only for these values of λ , and is a multiple of that polynomial. The following example explains why convergence only inside $(-1, 1)$ is not enough.

Axisymmetry: the ordinary Legendre equation (cont.)

Example. Take $\lambda = 2$ and start the *even* chain at $a_0 = 1$. Then (6.1.10) gives

$$a_2 = -\frac{2-0}{2 \cdot 1} = -1, \quad a_4 = -\frac{2-6}{4 \cdot 3}(-1) = -\frac{1}{3}, \quad a_6 = -\frac{2-20}{6 \cdot 5} \left(-\frac{1}{3}\right) = -\frac{1}{5},$$

and the pattern is $a_{2k} = -1/(2k-1)$ for $k \geq 1$. Subtracting the Taylor series for $\log(1+x)$ and $\log(1-x)$ gives $\sum_{k \geq 1} x^{2k-1}/(2k-1) = \frac{1}{2} \log[(1+x)/(1-x)]$ for $|x| < 1$. Hence

$$v(x) = 1 - \sum_{k \geq 1} \frac{x^{2k}}{2k-1} = 1 - x \sum_{k \geq 1} \frac{x^{2k-1}}{2k-1} = 1 - \frac{x}{2} \log \frac{1+x}{1-x}.$$

This is a genuine solution of (6.1.9) with $\lambda = 2$ (a one-line check by substitution), finite at every interior point, yet $v(x) \rightarrow -\infty$ as $x \rightarrow 1^-$: useless on the sphere. Meanwhile the *odd* chain with $\lambda = 2$ terminates immediately and gives the polynomial $v = x$. For the allowed value $\lambda = n(n+1)$, the recurrence factors as

$$a_{j+2} = -\frac{(n-j)(n+j+1)}{(j+2)(j+1)} a_j. \quad (6.1.11)$$

Axisymmetry: the ordinary Legendre equation (cont.)

At $j = n$ one has $a_{n+2} = 0$, so the same-parity chain is a polynomial of degree n . Set the initial coefficient of the opposite-parity chain to zero. The surviving polynomial is bounded at both poles.

With $\lambda = n(n+1)$, equations (6.1.8)–(6.1.9) become the ordinary Legendre equation

$$\frac{d}{dx}[(1-x^2)v'(x)] + n(n+1)v(x) = 0, \quad (6.1.12)$$

or

$$(1-x^2)v'' - 2xv' + n(n+1)v = 0. \quad (6.1.13)$$

Choose the scale so that $P_n(1) = 1$; the resulting polynomial is the *Legendre polynomial* P_n . Since the recurrence advances by two and only the terminating parity chain is retained, its powers are $x^n, x^{n-2}, \dots$. Thus

$$P_n(-x) = (-1)^n P_n(x).$$

Rodrigues' formula below supplies the normalization directly. The generating function below fixes all the normalized coefficients explicitly.

Table of contents

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- 5 Laguerre polynomials
- 6 Chebyshev polynomials
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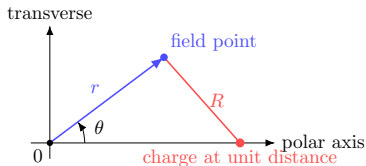
Generating function from the Coulomb potential

Step 1: write the axisymmetric potential. Place a point charge one unit from the origin on the positive polar axis. For a field point with spherical coordinates (r, θ, ϕ) , the law of cosines gives

$$R^2 = r^2 + 1^2 - 2r(1) \cos \theta = 1 - 2r \cos \theta + r^2,$$

so, for a nonzero physical scale factor C , the Coulomb potential is

$$\varphi(r, \theta) = \frac{C}{R} = \frac{C}{\sqrt{1 - 2r \cos \theta + r^2}}. \quad (6.2.1)$$



$$R^2 = 1 + r^2 - 2r \cos \theta$$

Generating function from the Coulomb potential (cont.)

In the picture the horizontal line is the polar axis $\theta = 0$, along which the charge sits, and θ is measured from it exactly as in (6.1.5). That axis is a genuine Cartesian direction; do not confuse it with the variable $x = \cos \theta$ used in the formulas of this chapter, which is the cosine of that angle and ranges over $[-1, 1]$.

For $r < 1$ the source is outside the region under study. The potential is harmonic, finite at the origin, and independent of ϕ . Our separated solutions therefore suggest the expansion

$$\frac{1}{\sqrt{1 - 2r \cos \theta + r^2}} = \sum_{n=0}^{\infty} a_n r^n P_n(\cos \theta). \quad (6.2.2)$$

This is a physical motivation for the generating function. We quote the existence of this convergent expansion for $r < 1$; the axis calculation now fixes its coefficients.

Step 2: determine the coefficients on the axis. Set $\theta = 0$. Then $\cos \theta = 1$, $P_n(1) = 1$, and

$$1 - 2r \cos 0 + r^2 = (1 - r)^2.$$

Because $0 \leq r < 1$, $|1 - r| = 1 - r$, so (6.2.2) becomes

$$\frac{1}{1 - r} = \sum_{n=0}^{\infty} a_n r^n = \sum_{n=0}^{\infty} r^n \quad (|r| < 1).$$

Generating function from the Coulomb potential (cont.)

Uniqueness of power-series coefficients gives $a_n = 1$. Writing $t = r$ and $x = \cos \theta$,

$$g(x, t) = \frac{1}{\sqrt{1 - 2xt + t^2}} = \sum_{n=0}^{\infty} P_n(x)t^n, \quad |t| < 1, \quad -1 \leq x \leq 1. \quad (6.2.3)$$

Step 3: expand the first three P_n from the generating function. For small t , write

$$g(x, t) = (1 - u)^{-1/2}, \quad u = 2xt - t^2,$$

and use the binomial series

$$(1 - u)^{-1/2} = 1 + \frac{1}{2}u + \frac{1}{2} \cdot \frac{3}{2} \frac{u^2}{2!} + O(u^3) = 1 + \frac{1}{2}u + \frac{3}{8}u^2 + O(u^3).$$

Collect powers of t up to t^2 :

$$\begin{aligned} u &= 2xt - t^2, \\ u^2 &= (2xt - t^2)^2 = 4x^2t^2 + O(t^3). \end{aligned}$$

Generating function from the Coulomb potential (cont.)

Hence

$$\begin{aligned}g &= 1 + \frac{1}{2}(2xt - t^2) + \frac{3}{8}(4x^2t^2) + O(t^3) \\&= 1 + xt + \left(-\frac{1}{2} + \frac{3}{2}x^2\right)t^2 + O(t^3) \\&= 1 + xt + \frac{1}{2}(3x^2 - 1)t^2 + O(t^3).\end{aligned}$$

Matching $\sum P_n(x)t^n$ therefore gives the first three polynomials explicitly:

$$P_0(x) = 1, \quad P_1(x) = x, \quad P_2(x) = \frac{1}{2}(3x^2 - 1). \quad (6.2.4)$$

Checks: $P_n(1) = 1$ and $P_n(-x) = (-1)^n P_n(x)$ hold for $n = 0, 1, 2$.

Three-term recurrence

Step 1: differentiate the generating function. Let

$$D(x, t) = 1 - 2xt + t^2, \quad g(x, t) = D(x, t)^{-1/2}.$$

Differentiating with respect to t gives

$$\frac{\partial g}{\partial t} = -\frac{1}{2}D^{-3/2}\frac{\partial D}{\partial t} = \frac{x-t}{D^{3/2}}.$$

Termwise differentiation of the power series gives

$$\frac{\partial g}{\partial t} = \sum_{n=1}^{\infty} nP_n(x)t^{n-1}. \quad (6.2.5)$$

Multiplying $g_t = (x-t)D^{-3/2}$ by D gives $(x-t)g = Dg_t$. Substituting both series,

$$(x-t)\sum_{n=0}^{\infty} P_nt^n = (1-2xt+t^2)\sum_{n=1}^{\infty} nP_nt^{n-1},$$

where $P_n = P_n(x)$. Expanding,

$$\sum_{n=0}^{\infty} xP_nt^n - \sum_{n=0}^{\infty} P_nt^{n+1} = \sum_{n=1}^{\infty} nP_nt^{n-1} - 2\sum_{n=1}^{\infty} n xP_nt^n + \sum_{n=1}^{\infty} nP_nt^{n+1}. \quad (6.2.6)$$

Three-term recurrence (cont.)

Step 2: compare coefficients of t^n . With the convention $P_{-1} = 0$, the left coefficient of t^n is $xP_n - P_{n-1}$. On the right,

$$[t^n] \sum_{j \geq 1} j P_j t^{j-1} = (n+1)P_{n+1}, \quad [t^n] \left(-2 \sum_{j \geq 1} j x P_j t^j \right) = -2n x P_n,$$

and

$$[t^n] \sum_{j \geq 1} j P_j t^{j+1} = (n-1)P_{n-1}$$

(using $P_{-1} = 0$). Equating coefficients,

$$xP_n - P_{n-1} = (n+1)P_{n+1} - 2n x P_n + (n-1)P_{n-1}.$$

Collecting the P_n and P_{n-1} terms gives

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x), \quad n \geq 0, \quad (6.2.7)$$

with $P_{-1} = 0$ and $P_0 = 1$. For $n = 0$, $P_1 = xP_0 = x$.

First few Legendre polynomials

We already obtained $P_0 = 1$, $P_1 = x$, and $P_2 = (3x^2 - 1)/2$ from the generating function. The recurrence (6.2.7) supplies the next one without another series expansion:

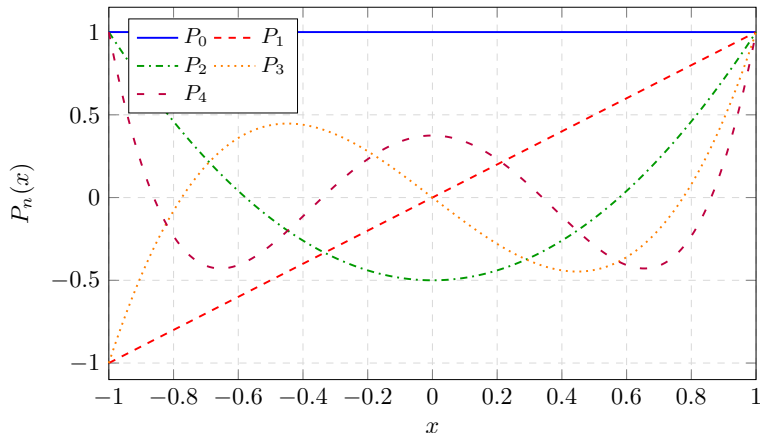
$$3P_3 = 5xP_2 - 2P_1 = \frac{5x(3x^2 - 1)}{2} - 2x = \frac{15x^3 - 9x}{2}.$$

Thus

$$P_3(x) = \frac{1}{2}(5x^3 - 3x).$$

It has odd parity and $P_3(1) = 1$, as expected. The following plot shows how the oscillations change with degree.

Shapes of the first Legendre polynomials



Each P_n is bounded on $[-1, 1]$ and normalized by $P_n(1) = 1$. Increasing n adds interior zeros while preserving $\int_{-1}^1 P_m(x)P_n(x) dx = 0$ for $m \neq n$.

Rodrigues' formula

The recurrence generates $P_0, P_1, P_2, \dots$ in order. A complementary formula constructs any one polynomial directly:

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n. \quad (6.2.8)$$

This is Rodrigues' formula. We use the identity without its general proof. For $n = 2$, the calculation is just polynomial differentiation:

$$\begin{aligned} (x^2 - 1)^2 &= x^4 - 2x^2 + 1, \\ \frac{d}{dx} (x^2 - 1)^2 &= 4x^3 - 4x, \\ \frac{d^2}{dx^2} (x^2 - 1)^2 &= 12x^2 - 4, \\ P_2(x) &= \frac{12x^2 - 4}{2^2 2!} = \frac{3x^2 - 1}{2}. \end{aligned}$$

The factor $2^n n!$ makes $P_n(1) = 1$: in $(x - 1)^n (x + 1)^n$, the only term surviving after n derivatives at $x = 1$ differentiates $(x - 1)^n$ all n times and equals $n! 2^n$.

Rodrigues' formula (cont.)

Two consequences needed for the norm. The highest power in $(x^2 - 1)^n$ is x^{2n} . Hence

$$\frac{d^n}{dx^n} x^{2n} = \frac{(2n)!}{n!} x^n,$$

and division by $2^n n!$ gives

$$P_n(x) = \frac{(2n)!}{2^n (n!)^2} x^n + \text{lower powers.} \quad (6.2.9)$$

Differentiating n more times removes all lower powers:

$$P_n^{(n)}(x) = \frac{(2n)!}{2^n n!}. \quad (6.2.10)$$

The zeros of order n in $(x^2 - 1)^n$ at $x = \pm 1$ also make the boundary terms vanish when we integrate by parts up to n times. These are the two facts used next.

Connection to Chapter 1. Cauchy's derivative formula applied to $(t^2 - 1)^n$ also writes Rodrigues' formula as

$$P_n(x) = \frac{1}{2^{n+1}\pi i} \oint_C \frac{(t^2 - 1)^n}{(t - x)^{n+1}} dt, \quad (6.2.11)$$

where C is a simple counterclockwise contour enclosing x . We will use the derivative form for the norm calculation.

Orthogonality and norm of Legendre polynomials

The Legendre equation is

$$\frac{d}{dx} [(1-x^2)P_n'(x)] + n(n+1)P_n(x) = 0. \quad (6.2.12)$$

This is the Chapter 4 problem with $p = 1 - x^2$, $q = 0$, and $w = 1$. At either endpoint,

$$(1-x^2)(P_n P_m' - P_n' P_m) \longrightarrow 0,$$

since polynomials and their derivatives remain finite. The eigenvalues $n(n+1)$ differ for distinct nonnegative degrees. Therefore Chapter 4's weighted orthogonality result gives

$$\int_{-1}^1 P_n(x) P_m(x) dx = 0 \quad (n \neq m). \quad (6.2.13)$$

Orthogonality and norm of Legendre polynomials (cont.)

Norm via Rodrigues and integration by parts. For $n = 0$, $P_0 = 1$ gives $I_0 = 2$ directly. For $n \geq 1$, set

$$I_n := \int_{-1}^1 P_n(x)^2 dx = \frac{1}{2^n n!} \int_{-1}^1 P_n(x) \frac{d^n}{dx^n} (x^2 - 1)^n dx.$$

Integrate by parts once:

$$\begin{aligned} I_n &= \frac{1}{2^n n!} \left[P_n \frac{d^{n-1}}{dx^{n-1}} (x^2 - 1)^n \right]_{-1}^1 \\ &\quad - \frac{1}{2^n n!} \int_{-1}^1 P'_n \frac{d^{n-1}}{dx^{n-1}} (x^2 - 1)^n dx. \end{aligned}$$

Because $(x^2 - 1)^n = (x - 1)^n (x + 1)^n$ has a zero of order n at each endpoint,

$$\left. \frac{d^j}{dx^j} (x^2 - 1)^n \right|_{x=\pm 1} = 0, \quad 0 \leq j \leq n - 1.$$

Orthogonality and norm of Legendre polynomials (cont.)

Thus every boundary term in n successive integrations by parts vanishes, and

$$I_n = \frac{(-1)^n}{2^n n!} \int_{-1}^1 P_n^{(n)}(x)(x^2 - 1)^n dx. \quad (6.2.14)$$

Use $P_n^{(n)} = (2n)!/(2^n n!)$ from (6.2.10) and $(-1)^n(x^2 - 1)^n = (1 - x^2)^n$:

$$I_n = \frac{(2n)!}{2^{2n}(n!)^2} \int_{-1}^1 (1 - x^2)^n dx.$$

Orthogonality and norm of Legendre polynomials (cont.)

To evaluate the remaining integral, set $x = 2u - 1$. Then $dx = 2 du$ and $1 - x^2 = 4u(1 - u)$, so with the beta function

$$B(a, b) = \int_0^1 u^{a-1}(1-u)^{b-1} du = \Gamma(a)\Gamma(b)/\Gamma(a+b),$$

$$\begin{aligned}\int_{-1}^1 (1-x^2)^n dx &= 2 \cdot 4^n \int_0^1 u^n (1-u)^n du = 2^{2n+1} B(n+1, n+1) \\ &= 2^{2n+1} \frac{(n!)^2}{(2n+1)!}.\end{aligned}$$

Therefore

$$I_n = \frac{(2n)!}{2^{2n}(n!)^2} \cdot \frac{2^{2n+1}(n!)^2}{(2n+1)!} = \frac{2}{2n+1}.$$

Combining orthogonality and norm,

$$\int_{-1}^1 P_n(x) P_m(x) dx = \frac{2}{2n+1} \delta_{nm}. \quad (6.2.15)$$

Fourier–Legendre series

We use the standard completeness result without proof: Legendre polynomials form a complete orthogonal system in $L^2([-1, 1])$, just as the sines did in Chapter 4. With projection coefficients, their partial sums converge in the L^2 norm. Thus, for $f \in L^2([-1, 1])$,

$$f(x) = \sum_{n=0}^{\infty} a_n P_n(x)$$

in the L^2 sense. Multiply by $P_m(x)$ and integrate:

$$\int_{-1}^1 f(x) P_m(x) dx = \sum_{n=0}^{\infty} a_n \int_{-1}^1 P_n(x) P_m(x) dx = a_m \frac{2}{2m+1}.$$

Solving for a_m gives

$$a_m = \frac{2m+1}{2} \int_{-1}^1 f(x) P_m(x) dx. \quad (6.2.16)$$

Therefore the Fourier–Legendre expansion is

$$f(x) = \sum_{n=0}^{\infty} \left[\frac{2n+1}{2} \int_{-1}^1 f(s) P_n(s) ds \right] P_n(x). \quad (6.2.17)$$

Useful formulas for the Legendre polynomial

differential equation $(1 - x^2)P_n'' - 2xP_n' + n(n+1)P_n = 0$

series solution
$$P_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(-1)^k (2n-2k)!}{2^n k! (n-2k)! (n-k)!} x^{n-2k}$$

generating function
$$\frac{1}{\sqrt{1-2xt+t^2}} = \sum_{n=0}^{\infty} P_n(x) t^n$$

recurrence $(n+1)P_{n+1} = (2n+1)xP_n - nP_{n-1}$

orthogonality
$$\int_{-1}^1 P_m(x)P_n(x) dx = \frac{2}{2n+1} \delta_{nm}$$

Rodrigues
$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$$

Schl\"afli
$$P_n(x) = \frac{1}{2^{n+1} \pi i} \oint \frac{(t^2 - 1)^n}{(t - x)^{n+1}} dt \quad (6.2.11)$$

The finite sum follows by expanding Rodrigues' formula; $\lfloor n/2 \rfloor$ means the greatest integer no larger than $n/2$. Generating series: $|t| < 1$ for $x \in [-1, 1]$. The contour encloses $t = x$.

Table of contents

- 1 From Laplace to ordinary Legendre
- 2 Legendre polynomials
- 3 Associated Legendre and spherical harmonics**
- 4 Hermite polynomials
- 5 Laguerre polynomials
- 6 Chebyshev polynomials
- 7 Summary

Associated Legendre equation and Y_n^m

When the field depends on ϕ , separate $y(\theta, \phi) = v(\theta)w(\phi)$. As in the polar calculation, the separated azimuthal modes can be chosen as $w = e^{im\phi}$; periodicity $w(\phi + 2\pi) = w(\phi)$ requires $m \in \mathbb{Z}$. Return to the angular equation in §6.1 and put $y(\theta, \phi) = v(\theta)e^{im\phi}$:

$$\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 y}{\partial \phi^2} + \lambda y = 0.$$

Since $\frac{\partial^2}{\partial \phi^2} e^{im\phi} = (im)^2 e^{im\phi} = -m^2 e^{im\phi}$, every term carries the common factor $e^{im\phi}$, which is never zero and therefore cancels:

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dv}{d\theta} \right) + \left[\lambda - \frac{m^2}{\sin^2 \theta} \right] v = 0.$$

Now set $x = \cos \theta$. The first term was already converted in the axisymmetric frame: there $\sin \theta \, dv/d\theta = -(1 - x^2)v'$ gave

$$\frac{1}{\sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{dv}{d\theta} \right) = \frac{d}{dx} [(1 - x^2)v'(x)],$$

Associated Legendre equation and Y_n^m (cont.)

and for the new term $\sin^2 \theta = 1 - \cos^2 \theta = 1 - x^2$, so $m^2 / \sin^2 \theta$ becomes $m^2 / (1 - x^2)$. The polar equation is therefore

$$\frac{d}{dx} [(1 - x^2)v'(x)] + \left[\lambda - \frac{m^2}{1 - x^2} \right] v(x) = 0. \quad (6.3.1)$$

For a smooth angular field on the whole sphere, the allowed values are $\lambda = n(n + 1)$ with $n = |m|, |m| + 1, \dots$ (endpoint result quoted). The equation becomes

$$(1 - x^2)v'' - 2xv' + \left[n(n + 1) - \frac{m^2}{1 - x^2} \right] v = 0. \quad (6.3.2)$$

Instead of repeating a long series analysis at the poles, construct these solutions from the Legendre polynomials already known.

Associated Legendre equation and Y_n^m (cont.)

A derivative construction (quoted). The associated Legendre functions on $[-1, 1]$ are obtained from the known polynomials by

$$P_n^m(x) = (-1)^m (1 - x^2)^{m/2} \frac{d^m P_n(x)}{dx^m}, \quad m = 0, 1, \dots, n. \quad (6.3.3)$$

The sign $(-1)^m$ is a convention, called the Condon–Shortley phase. For example, $P_1 = x$ and $P_2 = (3x^2 - 1)/2$ give

$$P_1^1(x) = -\sqrt{1 - x^2},$$

$$P_2^1(x) = -3x\sqrt{1 - x^2},$$

$$P_2^2(x) = (1 - x^2) \frac{d^2}{dx^2} \frac{3x^2 - 1}{2} = 3(1 - x^2).$$

For $m = 0$ this returns P_n . For $m > n$, the derivative is zero, so it produces no additional mode. This construction supplies the regular angular solutions selected above.

Convention warning. Much of the applied-scattering literature omits the Condon–Shortley factor, writing $P_n^m(x) = (1 - x^2)^{m/2} P_n^{(m)}(x)$; then $P_n^1(\cos \theta) = \sin \theta P_n'(\cos \theta)$ (prime in the argument $x = \cos \theta$), e.g. $P_1^1(\cos \theta) = \sin \theta$. Check the sign convention before combining formulas from different sources.

Associated Legendre equation and Y_n^m (cont.)

Spherical harmonics: name the angular modes. An unnormalized angular basis is

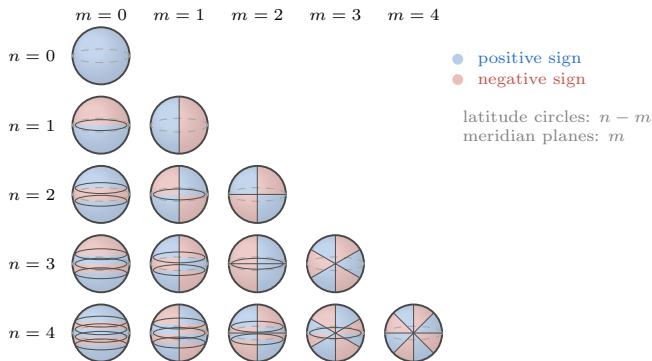
$$\mathcal{Y}_n^m(\theta, \phi) = P_n^{|m|}(\cos \theta) e^{im\phi}, \quad -n \leq m \leq n.$$

Normalized spherical harmonics Y_n^m differ from this by a constant and a fixed phase. Many texts call the degree ℓ and write the harmonics as $Y_{\ell m}$; here the degree is called n . With the same normalization and phase convention, $Y_{\ell m}$ and Y_n^m denote the same function when $\ell = n$. Combining radial and angular factors, the regular interior expansion is

$$\psi_{\text{int}}(r, \theta, \phi) = \sum_{n=0}^{\infty} \sum_{m=-n}^n a_{nm} r^n \mathcal{Y}_n^m(\theta, \phi), \quad (6.3.4)$$

and the decaying exterior expansion uses r^{-n-1} instead of r^n .

Spherical-harmonic nodal patterns



Schematic counts, not exact nodal locations. Rows: degree n ; columns: order m . The factor $P_n^{|m|}(\cos \theta)$ has $n - |m|$ interior nodal latitudes. For $m > 0$, a real factor $\cos(m\phi)$ or $\sin(m\phi)$ adds m meridian planes; for $m = 0$, there are none.

When the degree is not an integer

The Legendre equation makes sense with $\lambda = \nu(\nu + 1)$ for noninteger ν ; what integer n supplied was a solution regular at *both* endpoints $x = \pm 1$. Requiring regularity only at $x = 1$ leaves the *Legendre function of the first kind* P_ν , defined by the hypergeometric series

$$P_\nu(x) = {}_2F_1\left(-\nu, \nu + 1; 1; \frac{1-x}{2}\right), \quad (6.3.5)$$

where for complex α, β, γ with $\gamma \notin \{0, -1, -2, \dots\}$ and $|z| < 1$,

$${}_2F_1(\alpha, \beta; \gamma; z) = \sum_{j=0}^{\infty} \frac{(\alpha)_j(\beta)_j}{(\gamma)_j j!} z^j, \quad (\alpha)_0 = 1, \quad (\alpha)_j = \alpha(\alpha + 1) \cdots (\alpha + j - 1).$$

Consequences read off the series:

- $P_\nu(1) = 1$ for every degree—the argument $(1-x)/2$ vanishes at $x = 1$.
- At integer $\nu = n$, the factor $(-n)_j$ vanishes for $j > n$: the series terminates and (6.3.5) recovers P_n .
- $P_{-\nu-1} = P_\nu$, since the numerator parameters enter symmetrically.
- For noninteger degree, P_ν is logarithmically singular at $x = -1$ (quoted). Applications on $0 < \theta < \pi$ therefore use the reflected argument $-\cos \theta$, which reaches the regular point $+1$ at $\theta = \pi$.

When the degree is not an integer (cont.)

In wave problems the degree becomes a complex variable when a modal sum is rewritten as a contour integral in the order; $P_{\nu-1/2}(-\cos\theta)$ is then the angular factor sampled at integer and half-integer points. At $\nu = n + \frac{1}{2}$ the degree is the integer n , and $P_n(-\cos\theta) = (-1)^n P_n(\cos\theta)$ returns the polynomial case.

From scalar harmonics to tangential vector fields

Fields on a sphere can also be vectors. The vector harmonics are built from the scalar modes by curls: for a scalar radial-angular mode $\psi = z_n(kr) Y(\theta, \phi)$,

$$\vec{M} = \nabla \times (\vec{r} \psi), \quad \vec{N} = \frac{1}{k} \nabla \times \vec{M}.$$

Since $\vec{M} = \nabla \psi \times \vec{r}$ is perpendicular to $\hat{r}$, it is purely tangential—called *transverse electric to $\hat{r}$* (TE); $\vec{N}$ is its transverse-magnetic companion (TM). Both satisfy the vector Helmholtz equation when ψ satisfies the scalar one.

The point for this course: **no new special functions appear**. The angular factors are the associated Legendre modes $P_n^{|m|}(\cos \theta) e^{im\phi}$ and their θ -derivatives; the radial factors are the spherical Bessel and Hankel functions $j_n, h_n^{(1,2)}$ of Chapter 5, with matching conditions written through the Riccati forms $xz_n(x)$. That is the full toolkit a scattering course needs for the sphere.

Table of contents

- 1 From Laplace to ordinary Legendre
- 2 Legendre polynomials
- 3 Associated Legendre and spherical harmonics
- 4 Hermite polynomials**
- 5 Laguerre polynomials
- 6 Chebyshev polynomials
- 7 Summary

The Hermite equation and series solution

Step 1: substitute a power series. The physicists' Hermite polynomials solve

$$y'' - 2xy' + 2ny = 0, \quad n = 0, 1, 2, \dots \quad (6.4.1)$$

Let $y(x) = \sum_{j=0}^{\infty} a_j x^j$. Then

$$y' = \sum_{j=0}^{\infty} (j+1) a_{j+1} x^j,$$

$$y'' = \sum_{j=0}^{\infty} (j+2)(j+1) a_{j+2} x^j,$$

$$-2xy' = -2 \sum_{j=0}^{\infty} j a_j x^j.$$

Substitution into (6.4.1) gives

$$\sum_{j=0}^{\infty} [(j+2)(j+1)a_{j+2} + 2(n-j)a_j] x^j = 0.$$

The Hermite equation and series solution (cont.)

Every coefficient vanishes, so

$$a_{j+2} = \frac{2(j-n)}{(j+2)(j+1)} a_j. \quad (6.4.2)$$

Step 2: termination condition. Even and odd chains are independent. For integer $n \geq 0$, set $j = n$ in (6.4.2):

$$a_{n+2} = \frac{2(n-n)}{(n+2)(n+1)} a_n = 0.$$

Thus the same-parity chain terminates at degree n , giving a polynomial of exact degree n . The opposite-parity chain never hits a zero numerator, so we set its initial coefficient to zero (the non-polynomial second solution of (6.4.1)).

Step 3: explicit series from the downward product. Adopt the physicists' leading coefficient $a_n = 2^n$. The two-step recurrence (6.4.2) run *downward* from degree n is

$$a_j = \frac{(j+2)(j+1)}{2(j-n)} a_{j+2} \quad (j = n-2, n-4, \dots).$$

The Hermite equation and series solution (cont.)

The first two downward steps are

$$a_{n-2} = -\frac{n(n-1)}{4} a_n, \quad a_{n-4} = -\frac{(n-2)(n-3)}{8} a_{n-2},$$

whenever the indicated indices are nonnegative. After k steps, the numerator contains the first $2k$ descending factors of $n!$, and the denominator is $(-4)(-8)\cdots(-4k) = (-4)^k k!$. Hence

$$a_{n-2k} = \frac{2^n n!}{(-4)^k k! (n-2k)!} = \frac{(-1)^k 2^{n-2k} n!}{k! (n-2k)!}.$$

Writing $a_{n-2k} x^{n-2k} = (-1)^k (2x)^{n-2k} n! / (k! (n-2k)!)$ and summing gives

$$H_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k (2x)^{n-2k} \frac{n!}{k! (n-2k)!}. \quad (6.4.3)$$

Low- n checks:

$$H_0 = 1, \quad H_1 = 2x, \quad H_2 = 4x^2 - 2, \quad H_3 = 8x^3 - 12x.$$

The Hermite equation and series solution (cont.)

Multiplying (6.4.1) by e^{-x^2} yields the Sturm–Liouville form

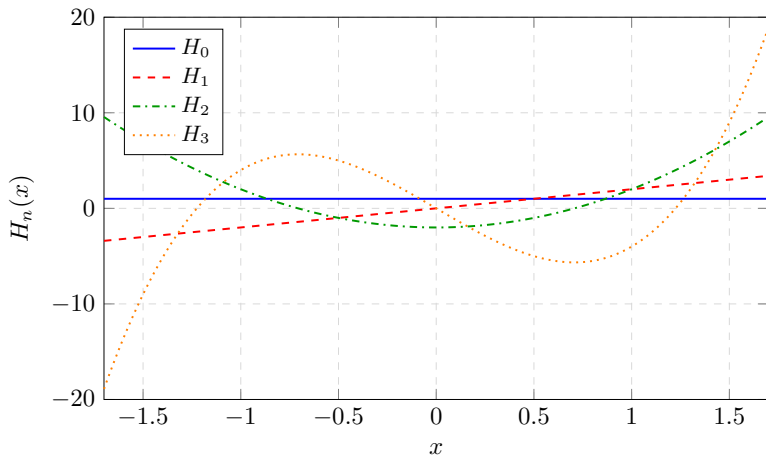
$$\frac{d}{dx} \left(e^{-x^2} y' \right) + 2n e^{-x^2} y = 0. \quad (6.4.4)$$

The same Lagrange-identity argument used for Legendre shows, for $n \neq m$,

$$\int_{-\infty}^{\infty} H_n(x) H_m(x) e^{-x^2} dx = 0. \quad (6.4.5)$$

(The diagonal norm is computed from the generating function below.)

Shapes of the first Hermite polynomials



Hermite polynomials grow away from the origin; the physical oscillator functions are $e^{-x^2/2}H_n(x)$.

Quantum harmonic oscillator

For a mass- M particle in $V(x) = \frac{1}{2}M\omega^2x^2$, the time-independent Schrödinger equation is given below. Here E is the energy, $\omega > 0$ is the oscillator's angular frequency, $\hbar$ is the reduced Planck constant, and ψ is the spatial wavefunction:

$$-\frac{\hbar^2}{2M}\psi'' + \frac{1}{2}M\omega^2x^2\psi = E\psi.$$

Use $\xi = \sqrt{M\omega/\hbar}x$. Then

$$\frac{d^2\psi}{dx^2} = \frac{M\omega}{\hbar} \frac{d^2\psi}{d\xi^2}, \quad \frac{1}{2}M\omega^2x^2 = \frac{1}{2}\hbar\omega\xi^2.$$

Substitution and division by $\hbar\omega/2$ give $-\psi_{\xi\xi} + \xi^2\psi = (2E/\hbar\omega)\psi$, or

$$\frac{d^2\psi}{d\xi^2} + (\varepsilon - \xi^2)\psi = 0, \quad \varepsilon = \frac{2E}{\hbar\omega}. \quad (6.4.6)$$

For large $|\xi|$, $\psi'' - \xi^2\psi \approx 0$ suggests the decaying factor $e^{-\xi^2/2}$. Set

$$\psi(\xi) = e^{-\xi^2/2}y(\xi).$$

Quantum harmonic oscillator (cont.)

Compute the derivatives by the product rule. Write $f := e^{-\xi^2/2}$, so $f' = -\xi f$ and $f'' = (-1 + \xi^2)f$ (because $f'' = -(f + \xi f') = -(f + \xi(-\xi f)) = (-1 + \xi^2)f$). Then

$$\begin{aligned}\psi' &= f'y + fy' = e^{-\xi^2/2}(y' - \xi y), \\ \psi'' &= f''y + 2f'y' + fy'' \\ &= e^{-\xi^2/2} \left[(-1 + \xi^2)y + 2(-\xi)y' + y'' \right] \\ &= e^{-\xi^2/2} (y'' - 2\xi y' + (\xi^2 - 1)y).\end{aligned}$$

Substitute into (6.4.6):

$$\psi'' + (\varepsilon - \xi^2)\psi = e^{-\xi^2/2} \left[y'' - 2\xi y' + (\xi^2 - 1)y + (\varepsilon - \xi^2)y \right] = e^{-\xi^2/2} [y'' - 2\xi y' + (\varepsilon - 1)y].$$

The Gaussian prefactor never vanishes, so

$$y'' - 2\xi y' + (\varepsilon - 1)y = 0. \tag{6.4.7}$$

A power series $y = \sum a_j \xi^j$ has coefficient recurrence

$$a_{j+2} = \frac{2j + 1 - \varepsilon}{(j+2)(j+1)} a_j.$$

Quantum harmonic oscillator (cont.)

Why choose a terminating series? A bound state requires $\int_{-\infty}^{\infty} |\psi|^2 d\xi < \infty$. A polynomial y times $e^{-\xi^2/2}$ satisfies this requirement. In contrast, a nonterminating parity chain has

$$\frac{a_{j+2}}{a_j} \sim \frac{2}{j},$$

like the coefficients of e^{ξ^2} . This suggests a growing exponential component that the prefactor $e^{-\xi^2/2}$ cannot suppress. This coefficient comparison is intuition, not a full asymptotic proof. We use the standard bound-state result: decay at both ends selects a terminating chain, with the other parity chain absent.

Termination at degree n means the numerator $2j + 1 - \varepsilon$ vanishes at $j = n$, i.e.

$$\varepsilon = 2n + 1, \quad n = 0, 1, 2, \dots,$$

and then $\varepsilon - 1 = 2n$, so (6.4.7) is exactly the Hermite equation $y'' - 2\xi y' + 2ny = 0$. Since $\varepsilon = 2E/(\hbar\omega)$, the energies are therefore

$$E_n = \left(n + \frac{1}{2}\right) \hbar\omega, \quad n = 0, 1, 2, \dots, \quad (6.4.8)$$

and the spatial factors are $\psi_n \propto e^{-\xi^2/2} H_n(\xi)$.

Generating function, norm, and Rodrigues

Step 1: generating function from the series. Define

$$g(x, t) := \exp(2xt - t^2) = e^{2xt} e^{-t^2}.$$

Expand each exponential:

$$e^{2xt} = \sum_{m=0}^{\infty} \frac{(2x)^m}{m!} t^m, \quad e^{-t^2} = \sum_{j=0}^{\infty} \frac{(-1)^j}{j!} t^{2j}.$$

The product is a double sum. The coefficient of t^n arises from pairs (m, j) with $m + 2j = n$, i.e. $m = n - 2j$ for $j = 0, \dots, \lfloor n/2 \rfloor$:

$$\begin{aligned} [t^n]g(x, t) &= \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(2x)^{n-2j}}{(n-2j)!} \cdot \frac{(-1)^j}{j!} \\ &= \frac{1}{n!} \sum_{j=0}^{\lfloor n/2 \rfloor} (-1)^j (2x)^{n-2j} \frac{n!}{j! (n-2j)!}. \end{aligned}$$

Generating function, norm, and Rodrigues (cont.)

Comparing with the series solution (6.4.3), the sum in brackets is exactly $H_n(x)$.
Therefore

$$g(x, t) = e^{-t^2+2xt} = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!}. \quad (6.4.9)$$

Step 2: Rodrigues from $g = e^{x^2} e^{-(x-t)^2}$. Rewrite the exponent:

$$2xt - t^2 = x^2 - (x - t)^2, \quad g(x, t) = e^{x^2} e^{-(x-t)^2}.$$

Taylor-expand in t about $t = 0$:

$$e^{-(x-t)^2} = \sum_{n=0}^{\infty} \frac{t^n}{n!} \left. \frac{\partial^n}{\partial t^n} e^{-(x-t)^2} \right|_{t=0}.$$

On the composite $e^{-(x-t)^2}$, one has $\partial/\partial t = -\partial/\partial x$, so

$$\left. \frac{\partial^n}{\partial t^n} e^{-(x-t)^2} \right|_{t=0} = (-1)^n \frac{d^n}{dx^n} e^{-x^2}.$$

Generating function, norm, and Rodrigues (cont.)

Hence

$$g(x, t) = e^{x^2} \sum_{n=0}^{\infty} \frac{t^n}{n!} (-1)^n \frac{d^n}{dx^n} e^{-x^2} = \sum_{n=0}^{\infty} \left[(-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2} \right] \frac{t^n}{n!}.$$

Matching coefficients with (6.4.9) gives Rodrigues' formula

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}. \quad (6.4.10)$$

Step 3: recurrences from $\partial_t g$ and $\partial_x g$. Differentiate the closed form:

$$\frac{\partial g}{\partial t} = (2x - 2t)g, \quad \frac{\partial g}{\partial x} = 2tg.$$

Using the series (6.4.9),

$$\partial_t g = \sum_{n=1}^{\infty} H_n \frac{t^{n-1}}{(n-1)!} = \sum_{n=0}^{\infty} H_{n+1} \frac{t^n}{n!}.$$

Multiplication by t shifts the index in the other term:

$$2tg = \sum_{n=1}^{\infty} 2H_{n-1} \frac{t^n}{(n-1)!} = \sum_{n=1}^{\infty} 2nH_{n-1} \frac{t^n}{n!},$$

Generating function, norm, and Rodrigues (cont.)

where $1/(n-1)! = n/n!$. Thus the coefficient of $t^n/n!$ in $(2x - 2t)g$ is $2xH_n - 2nH_{n-1}$. Comparing with $\partial_t g$ gives, for $n \geq 1$,

$$H_{n+1}(x) = 2xH_n(x) - 2nH_{n-1}(x). \quad (6.4.11)$$

Likewise, $\partial_x g = 2tg$. The same power shift just used gives

$$\sum_{n=0}^{\infty} H'_n(x) \frac{t^n}{n!} = \sum_{n=1}^{\infty} 2nH_{n-1}(x) \frac{t^n}{n!}.$$

Comparing coefficients of $t^n/n!$ gives

$$H'_n(x) = 2nH_{n-1}(x), \quad n \geq 1. \quad (6.4.12)$$

The constant coefficient separately gives $H'_0 = 0$.

Generating function, norm, and Rodrigues (cont.)

Step 4: norm via the double generating function. For real s, t near 0,

$$I(s, t) := \int_{-\infty}^{\infty} e^{-x^2} g(x, s) g(x, t) dx = \int_{-\infty}^{\infty} e^{-x^2 - s^2 + 2sx - t^2 + 2tx} dx.$$

Complete the square in x :

$$-x^2 + 2(s+t)x - s^2 - t^2 = -[x - (s+t)]^2 + 2st,$$

so

$$I(s, t) = \sqrt{\pi} e^{2st} = \sqrt{\pi} \sum_{k=0}^{\infty} \frac{(2st)^k}{k!} = \sqrt{\pi} \sum_{k=0}^{\infty} \frac{2^k}{k!} s^k t^k.$$

On the other hand, expand both generating functions:

$$g(x, s) g(x, t) = \sum_{m,n=0}^{\infty} H_m(x) H_n(x) \frac{s^m t^n}{m! n!},$$

For s, t near zero, the Gaussian decay permits termwise integration:

$$I(s, t) = \sum_{m,n=0}^{\infty} \left(\int_{-\infty}^{\infty} H_m(x) H_n(x) e^{-x^2} dx \right) \frac{s^m t^n}{m! n!}.$$

Generating function, norm, and Rodrigues (cont.)

The right-hand closed form $\sqrt{\pi} e^{2st}$ contains only diagonal powers $s^k t^k$. Equating coefficients of $s^m t^n$:

- if $m \neq n$, the right side has coefficient 0, so $\int H_m H_n e^{-x^2} dx = 0$;
- if $m = n$, compare $[s^n t^n]$:

$$\frac{1}{(n!)^2} \int_{-\infty}^{\infty} H_n^2 e^{-x^2} dx = \sqrt{\pi} \frac{2^n}{n!},$$

Multiply by $(n!)^2$ to obtain the diagonal norm.

Combining both cases,

$$\int_{-\infty}^{\infty} H_m(x) H_n(x) e^{-x^2} dx = 2^n n! \sqrt{\pi} \delta_{mn}. \quad (6.4.13)$$

Low- n check: $H_0 = 1$ and $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$.

Useful formulas for the Hermite polynomial

differential equation $H_n'' - 2x H_n' + 2n H_n = 0$

series solution
$$H_n(x) = \sum_{k=0}^{\lfloor n/2 \rfloor} (-1)^k (2x)^{n-2k} \frac{n!}{k! (n-2k)!}$$

generating function
$$e^{-t^2+2tx} = \sum_{n=0}^{\infty} H_n(x) \frac{t^n}{n!}$$

recurrence $H_n' = 2n H_{n-1}, \quad H_{n+1} = 2x H_n - 2n H_{n-1}$

orthogonality
$$\int_{-\infty}^{\infty} H_m H_n e^{-x^2} dx = n! 2^n \sqrt{\pi} \delta_{mn}$$

Rodrigues
$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2}$$

Here $n \geq 0$; the displayed recurrences use $n \geq 1$, with $H_0 = 1$, $H_1 = 2x$.

Table of contents

- 1 From Laplace to ordinary Legendre
- 2 Legendre polynomials
- 3 Associated Legendre and spherical harmonics
- 4 Hermite polynomials
- 5 Laguerre polynomials**
- 6 Chebyshev polynomials
- 7 Summary

Laguerre equation, polynomials, and orthogonality

Standard Laguerre polynomials solve

$$x y'' + (1 - x)y' + n y = 0, \quad n = 0, 1, 2, \dots \quad (6.5.1)$$

Generalized Laguerre polynomials occur in the radial hydrogen-atom equation. Here we study only the standard family, defined by (6.5.1) and normalized by $L_n(0) = 1$.

Step 1: power series and coefficient recurrence. Dividing by x gives $P = (1 - x)/x$ and $Q = n/x$, so $\lim_{x \rightarrow 0} xP = 1$ and $\lim_{x \rightarrow 0} x^2Q = 0$. The Chapter 3 indicial test gives $r(r - 1) + r = r^2 = 0$. The finite branch is therefore an ordinary power series; the second solution has a logarithmic singularity. We keep the branch regular at the origin. The polynomials grow at infinity, but the weight e^{-x} makes their squared norms finite.

Laguerre equation, polynomials, and orthogonality (cont.)

Set $y = \sum_{k=0}^{\infty} a_k x^k$. Then

$$y' = \sum_{k=0}^{\infty} (k+1) a_{k+1} x^k,$$

$$xy'' = \sum_{k=0}^{\infty} k(k+1) a_{k+1} x^k$$

$$\left(\text{from } y'' = \sum (k+2)(k+1) a_{k+2} x^k, \text{ times } x, \text{ reindex} \right),$$

$$-xy' = -\sum_{k=0}^{\infty} k a_k x^k,$$

$$ny = \sum_{k=0}^{\infty} n a_k x^k.$$

Collecting the coefficient of x^k in $xy'' + (1-x)y' + ny$ gives

$$k(k+1)a_{k+1} + (k+1)a_{k+1} + (n-k)a_k = (k+1)^2 a_{k+1} + (n-k)a_k.$$

Laguerre equation, polynomials, and orthogonality (cont.)

Setting every coefficient to zero yields the one-step recurrence

$$a_{k+1} = -\frac{n-k}{(k+1)^2} a_k, \quad k = 0, 1, 2, \dots \quad (6.5.2)$$

At $k = n$ one has $a_{n+1} = 0$, so the series terminates as a polynomial of degree n . With the conventional choice $a_0 = 1$, iterating (6.5.2) gives

$$a_k = (-1)^k \frac{n(n-1) \cdots (n-k+1)}{(k!)^2} = (-1)^k \frac{1}{k!} \binom{n}{k},$$

and therefore

$$L_n(x) = \sum_{k=0}^n (-1)^k \binom{n}{k} \frac{x^k}{k!}. \quad (6.5.3)$$

The first few:

$$L_0 = 1, \quad L_1 = 1 - x, \quad L_2 = 1 - 2x + \frac{1}{2}x^2, \quad L_3 = 1 - 3x + \frac{3}{2}x^2 - \frac{1}{6}x^3.$$

Step 2: Sturm–Liouville form and off-diagonal orthogonality. Multiplying (6.5.1) by e^{-x} produces the SL form $(xe^{-x}y')' + ne^{-x}y = 0$. The same boundary-vanishing

Laguerre equation, polynomials, and orthogonality (cont.)

argument as for Hermite (polynomials and their derivatives are finite at zero, and e^{-x} dominates their growth at infinity) gives, for $n \neq m$,

$$\int_0^\infty L_n(x)L_m(x)e^{-x} dx = 0. \quad (6.5.4)$$

Step 3: generating function and the diagonal norm. Define

$$g(x, t) := \frac{1}{1-t} \exp\left(-\frac{xt}{1-t}\right) = \sum_{n=0}^{\infty} L_n(x) t^n, \quad |t| < 1. \quad (6.5.5)$$

The first equality is the definition of g ; the second is a claim about the polynomials (6.5.3), and we prove it now. Expand the exponential and absorb the prefactor:

$$\frac{1}{1-t} \exp\left(-\frac{xt}{1-t}\right) = \frac{1}{1-t} \sum_{k=0}^{\infty} \frac{1}{k!} \left(\frac{-xt}{1-t}\right)^k = \sum_{k=0}^{\infty} \frac{(-x)^k}{k!} \frac{t^k}{(1-t)^{k+1}}.$$

The binomial series from Chapter 1 gives

$$(1-t)^{-(k+1)} = \sum_{j=0}^{\infty} \binom{k+j}{k} t^j, \quad |t| < 1.$$

Laguerre equation, polynomials, and orthogonality (cont.)

So the power t^n is produced by the pairs with $j = n - k$ and $0 \leq k \leq n$, each carrying the factor $\binom{k+(n-k)}{k} = \binom{n}{k}$:

$$[t^n]g(x, t) = \sum_{k=0}^n \frac{(-x)^k}{k!} \binom{n}{k} = \sum_{k=0}^n (-1)^k \binom{n}{k} \frac{x^k}{k!} = L_n(x)$$

by (6.5.3). The power series is absolutely convergent for $|t| < 1$, which permits the rearrangement.

Use the double-generating-function method just applied to Hermite. For real s, t near zero, put

$$\alpha = 1 + \frac{s}{1-s} + \frac{t}{1-t} = \frac{1-st}{(1-s)(1-t)} > 0.$$

Then

$$e^{-x}g(x, s)g(x, t) = \frac{e^{-\alpha x}}{(1-s)(1-t)}.$$

Since $\int_0^\infty e^{-\alpha x} dx = 1/\alpha$, the integral becomes

$$\int_0^\infty e^{-x}g(x, s)g(x, t) dx = \frac{1}{(1-s)(1-t)\alpha} = \frac{1}{1-st}.$$

Laguerre equation, polynomials, and orthogonality (cont.)

Near $s = t = 0$, the exponential decay permits term-by-term integration. Expand both sides. The right side is

$$\frac{1}{1-st} = \sum_{n=0}^{\infty} (st)^n = \sum_{n=0}^{\infty} s^n t^n.$$

The left side is

$$\sum_{m,n=0}^{\infty} \left(\int_0^{\infty} L_m(x) L_n(x) e^{-x} dx \right) s^m t^n.$$

Equating coefficients of $s^m t^n$: if $m \neq n$ the integral vanishes (already known); if $m = n$ the coefficient is 1. Therefore the family is orthonormal:

$$\int_0^{\infty} L_m(x) L_n(x) e^{-x} dx = \delta_{mn}. \quad (6.5.6)$$

Step 4: three-term recurrence from $\partial_t g$. From the closed form (6.5.5),

$$\frac{\partial g}{\partial t} = g \cdot \frac{1-t-x}{(1-t)^2},$$

Laguerre equation, polynomials, and orthogonality (cont.)

or equivalently

$$(1-t)^2 \partial_t g = (1-t-x)g. \quad (6.5.7)$$

Expand both sides in powers of t . With $g = \sum_{n \geq 0} L_n t^n$ one has $\partial_t g = \sum_{n \geq 0} (n+1)L_{n+1}t^n$, so the left side of (6.5.7) is

$$(1-2t+t^2) \sum_{n \geq 0} (n+1)L_{n+1}t^n.$$

The coefficient of t^n on the left is

$$(n+1)L_{n+1} - 2nL_n + (n-1)L_{n-1}$$

(for $n \geq 1$, with the convention $L_{-1} := 0$). On the right, $(1-t-x) \sum L_n t^n$ contributes

$$(1-x)L_n - L_{n-1}.$$

Equating coefficients and rearranging yields

$$(n+1)L_{n+1}(x) = (2n+1-x)L_n(x) - nL_{n-1}(x). \quad (6.5.8)$$

Laguerre equation, polynomials, and orthogonality (cont.)

Check at $n = 1$: $(2)L_2 = (3 - x)L_1 - L_0$ gives $2(1 - 2x + \frac{1}{2}x^2) = (3 - x)(1 - x) - 1$, which holds identically.

Step 5: Rodrigues formula. The derivative construction is

$$L_n(x) = \frac{e^x}{n!} \frac{d^n}{dx^n} (x^n e^{-x}). \quad (6.5.9)$$

For $n = 2$, two product-rule steps give

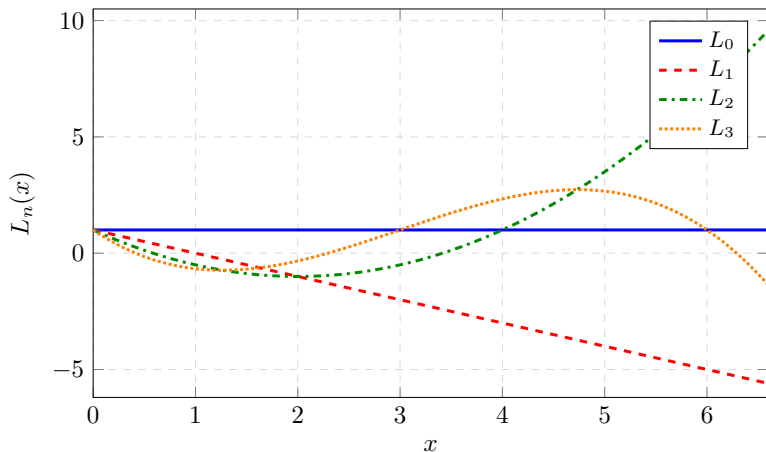
$$\begin{aligned} \frac{d}{dx}(x^2 e^{-x}) &= (2x - x^2)e^{-x}, \\ \frac{d^2}{dx^2}(x^2 e^{-x}) &= (2 - 4x + x^2)e^{-x}, \\ L_2(x) &= \frac{1}{2}(2 - 4x + x^2) = 1 - 2x + \frac{x^2}{2}. \end{aligned}$$

For general n , the product rule distributes k derivatives to e^{-x} and $n - k$ to x^n , giving

$$\frac{e^x}{n!} \sum_{k=0}^n \binom{n}{k} \frac{n!}{k!} x^k (-1)^k e^{-x} = \sum_{k=0}^n (-1)^k \binom{n}{k} \frac{x^k}{k!} = L_n(x).$$

Thus the derivative formula reproduces exactly the series already obtained.

Shapes of the first Laguerre polynomials



All standard Laguerre polynomials satisfy $L_n(0) = 1$. The weight e^{-x} makes them natural on $[0, \infty)$.

Useful formulas for the Laguerre polynomial

differential equation $xL_n'' + (1-x)L_n' + nL_n = 0$

series / first polynomials
$$L_n(x) = \sum_{k=0}^n (-1)^k \binom{n}{k} \frac{x^k}{k!}$$

$$L_0 = 1, L_1 = 1 - x, L_2 = 1 - 2x + \frac{1}{2}x^2$$

generating function
$$\frac{1}{1-t} \exp\left(-\frac{xt}{1-t}\right) = \sum_{n=0}^{\infty} L_n(x) t^n$$

recurrence
$$(n+1)L_{n+1} = (2n+1-x)L_n - nL_{n-1}$$

orthonormality
$$\int_0^{\infty} L_m L_n e^{-x} dx = \delta_{mn}$$

Rodrigues
$$L_n = \frac{e^x}{n!} \frac{d^n}{dx^n} (x^n e^{-x})$$

Generating series requires $|t| < 1$.

Table of contents

- 1 From Laplace to ordinary Legendre
- 2 Legendre polynomials
- 3 Associated Legendre and spherical harmonics
- 4 Hermite polynomials
- 5 Laguerre polynomials
- 6 Chebyshev polynomials**
- 7 Summary

Definition through $\cos n\theta$

The Chebyshev polynomial of the first kind is defined by the elementary identity

$$T_n(\cos \theta) = \cos(n\theta). \quad (6.6.1)$$

That is, set $x = \cos \theta$ and read off a polynomial in x :

$$T_0(x) = 1, \quad T_1(x) = x, \quad T_2(x) = 2x^2 - 1, \quad T_3(x) = 4x^3 - 3x.$$

(For T_2 : $\cos 2\theta = 2\cos^2 \theta - 1$. For T_3 : $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$.)

Recurrence from angle addition. Adding $\cos((n+1)\theta) = \cos(n\theta)\cos\theta - \sin(n\theta)\sin\theta$ and $\cos((n-1)\theta) = \cos(n\theta)\cos\theta + \sin(n\theta)\sin\theta$ cancels the sines:

$$T_{n+1}(x) + T_{n-1}(x) = 2xT_n(x).$$

Thus

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x), \quad n \geq 1, \quad (6.6.2)$$

with $T_0 = 1$, $T_1 = x$. Each step raises the degree by one: the leading term of $2xT_n$ cannot cancel with T_{n-1} . Starting with leading coefficient 1 for T_1 , it doubles at every step, so T_n has leading coefficient 2^{n-1} for $n \geq 1$.

Definition through $\cos n\theta$ (cont.)

The endpoints follow by setting $\theta = 0, \pi$: $T_n(1) = 1$ and $T_n(-1) = (-1)^n$. Replacing θ by $\pi - \theta$ changes x to $-x$, so

$$T_n(-x) = \cos(n\pi - n\theta) = (-1)^n T_n(x).$$

Generating function from the recurrence. A single power series packages all the T_n at once. Put

$$G(x, t) := \sum_{n=0}^{\infty} T_n(x) t^n.$$

For $x \in [-1, 1]$, $|T_n(x)| \leq 1$, so comparison with $\sum |t|^n$ allows regrouping when $|t| < 1$. Multiply (6.6.2) by t^{n+1} and sum over $n \geq 1$, which is the full range in which the recurrence holds:

$$\sum_{n=1}^{\infty} T_{n+1} t^{n+1} = 2xt \sum_{n=1}^{\infty} T_n t^n - t^2 \sum_{n=1}^{\infty} T_{n-1} t^{n-1}.$$

Definition through $\cos n\theta$ (cont.)

Each sum is G with a few terms removed or added. Using $T_0 = 1$ and $T_1 = x$,

$$\begin{aligned}\sum_{n=1}^{\infty} T_{n+1} t^{n+1} &= \sum_{k=2}^{\infty} T_k t^k = G - 1 - xt, & \sum_{n=1}^{\infty} T_n t^n &= G - 1, \\ \sum_{n=1}^{\infty} T_{n-1} t^{n-1} &= \sum_{k=0}^{\infty} T_k t^k = G.\end{aligned}$$

So $G - 1 - xt = 2xt(G - 1) - t^2 G$; collecting the G terms on the left and the rest on the right gives $(1 - 2xt + t^2)G = 1 + xt - 2xt = 1 - xt$, that is

$$\sum_{n=0}^{\infty} T_n(x) t^n = \frac{1 - xt}{1 - 2xt + t^2}, \quad |t| < 1, \quad -1 \leq x \leq 1. \quad (6.6.3)$$

Differential equation via the chain rule. Set $x = \cos \theta$ and $Y(\theta) := T_n(\cos \theta) = \cos(n\theta)$, so $Y''(\theta) = -n^2 \cos(n\theta) = -n^2 Y(\theta)$. Chain rule with $dx/d\theta = -\sin \theta$ gives

$$\frac{d}{d\theta} = -\sin \theta \frac{d}{dx}, \quad \frac{dY}{d\theta} = -\sin \theta T'_n(x).$$

Definition through $\cos n\theta$ (cont.)

Differentiate again:

$$\begin{aligned}\frac{d^2 Y}{d\theta^2} &= \frac{d}{d\theta} (-\sin \theta T'_n(x)) \\ &= -\cos \theta T'_n(x) - \sin \theta \cdot (-\sin \theta T''_n(x)) \\ &= -\cos \theta T'_n(x) + \sin^2 \theta T''_n(x).\end{aligned}$$

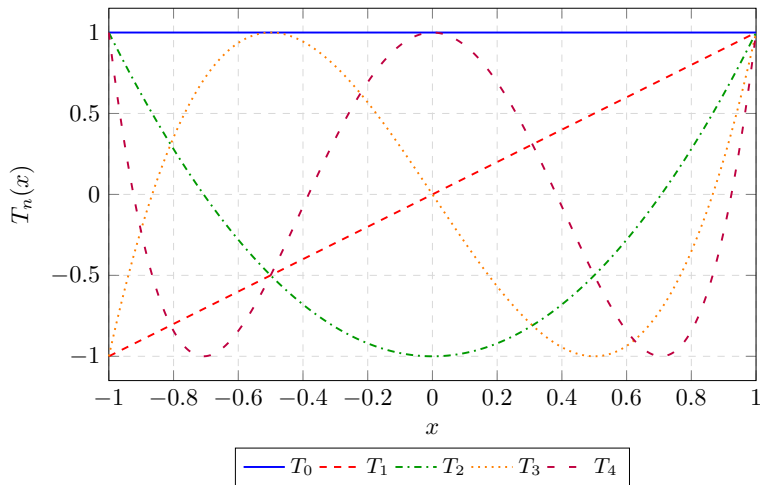
Substitute $\cos \theta = x$ and $\sin^2 \theta = 1 - x^2$, and use $Y'' = -n^2 Y = -n^2 T_n$:

$$(1 - x^2) T''_n(x) - x T'_n(x) = -n^2 T_n(x).$$

Rearranging yields the Chebyshev equation

$$(1 - x^2) T''_n(x) - x T'_n(x) + n^2 T_n(x) = 0. \quad (6.6.4)$$

Shapes of the first Chebyshev polynomials



$T_n(\cos \theta) = \cos(n\theta)$ keeps every T_n between -1 and 1 on $[-1, 1]$.

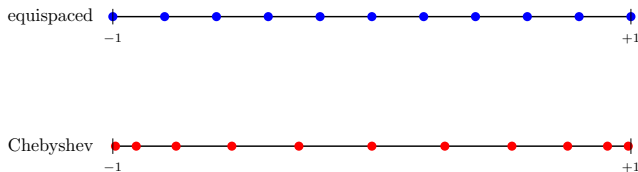
Chebyshev nodes and orthogonality

Zeros (nodes). For $n \geq 1$ and $x = \cos \theta \in [-1, 1]$,

$$T_n(x) = 0 \iff \cos(n\theta) = 0 \iff \theta_k = \frac{(2k-1)\pi}{2n}, \quad k = 1, \dots, n.$$

$$x_k = \cos\left(\frac{(2k-1)\pi}{2n}\right), \quad k = 1, \dots, n. \quad (6.6.5)$$

The angles are equally spaced, but $x = \cos \theta$ crowds the nodes near ± 1 . This endpoint clustering is useful for polynomial interpolation.



Chebyshev nodes and orthogonality (cont.)

Orthogonality. Dividing the differential equation by $\sqrt{1-x^2}$ gives

$$\left(\sqrt{1-x^2} T_n'\right)' + \frac{n^2}{\sqrt{1-x^2}} T_n = 0.$$

Thus the natural weight is $w(x) = 1/\sqrt{1-x^2}$ (up to a constant factor). More intuitively, this weight cancels the change-of-variable factor in $x = \cos \theta$, turning the integral into ordinary cosine orthogonality.

With the weight $(1-x^2)^{-1/2}$, substitute $x = \cos \theta$. Then $dx = -\sin \theta d\theta$, and as x runs from -1 to 1 , θ runs from π down to 0 . Also $\sqrt{1-x^2} = \sqrt{1-\cos^2 \theta} = \sin \theta$ for $\theta \in [0, \pi]$. Therefore

$$\begin{aligned} \int_{-1}^1 \frac{T_n(x) T_m(x)}{\sqrt{1-x^2}} dx &= \int_{\pi}^0 \frac{T_n(\cos \theta) T_m(\cos \theta)}{\sin \theta} (-\sin \theta) d\theta \\ &= \int_0^{\pi} T_n(\cos \theta) T_m(\cos \theta) d\theta \\ &= \int_0^{\pi} \cos(n\theta) \cos(m\theta) d\theta, \end{aligned}$$

Chebyshev nodes and orthogonality (cont.)

where the last step is the defining identity $T_k(\cos \theta) = \cos(k\theta)$. For $n \neq m$, use $\cos A \cos B = [\cos(A - B) + \cos(A + B)]/2$:

$$\int_0^\pi \cos(n\theta) \cos(m\theta) d\theta = \frac{1}{2} \left[\frac{\sin((n-m)\theta)}{n-m} + \frac{\sin((n+m)\theta)}{n+m} \right]_0^\pi = 0.$$

For $n = m \geq 1$, use $\cos^2(n\theta) = [1 + \cos(2n\theta)]/2$:

$$\int_0^\pi \cos^2(n\theta) d\theta = \left[\frac{\theta}{2} + \frac{\sin(2n\theta)}{4n} \right]_0^\pi = \frac{\pi}{2}.$$

For $n = m = 0$, the integrand is 1 and the integral is π . Hence

$$\int_{-1}^1 \frac{T_n(x) T_m(x)}{\sqrt{1-x^2}} dx = \begin{cases} \pi, & n = m = 0, \\ \pi/2, & n = m \geq 1, \\ 0, & n \neq m. \end{cases} \quad (6.6.6)$$

Useful formulas for the Chebyshev polynomial

definition	$T_n(\cos \theta) = \cos(n\theta)$
differential equation	$(1 - x^2)T_n'' - xT_n' + n^2T_n = 0$
first polynomials	$T_0 = 1, T_1 = x, T_2 = 2x^2 - 1, T_3 = 4x^3 - 3x$
recurrence	$T_{n+1} = 2xT_n - T_{n-1}$
generating function	$\frac{1 - xt}{1 - 2xt + t^2} = \sum_{n=0}^{\infty} T_n(x)t^n, \quad (6.6.3)$
zeros	$x_k = \cos\left(\frac{(2k-1)\pi}{2n}\right)$
orthogonality	$\int_{-1}^1 \frac{T_n T_m}{\sqrt{1-x^2}} dx = \begin{cases} \pi, & n = m = 0, \\ \pi/2, & n = m \geq 1, \\ 0, & n \neq m. \end{cases}$

Generating series requires $|t| < 1$.

Table of contents

- 1 From Laplace to ordinary Legendre
- 2 Legendre polynomials
- 3 Associated Legendre and spherical harmonics
- 4 Hermite polynomials
- 5 Laguerre polynomials
- 6 Chebyshev polynomials
- 7 Summary**

Summary: four orthogonal families

Family	Interval	Weight	ODE (sketch)
Legendre P_n	$[-1, 1]$	1	$(1 - x^2)y'' - 2xy' + n(n + 1)y = 0$
Hermite H_n	$(-\infty, \infty)$	e^{-x^2}	$y'' - 2xy' + 2ny = 0$
Laguerre L_n	$[0, \infty)$	e^{-x}	$xy'' + (1 - x)y' + ny = 0$
Chebyshev T_n	$[-1, 1]$	$(1 - x^2)^{-1/2}$	$(1 - x^2)y'' - xy' + n^2y = 0$

Physical origins (one line each). P_n : axisymmetric Laplace / multipoles; H_n : quantum harmonic oscillator; L_n : its generalized family occurs in the hydrogen radial equation; T_n : $\cos n\theta$ and stable interpolation.

The shared structure is a second-order Sturm–Liouville ODE, a recurrence, and an orthogonality relation on a natural interval and weight. Orthogonality alone only tells us how to *compute* the coefficients of a series; to know that the series actually reproduces f we need the separate property of *completeness*, stated for P_n on the Fourier–Legendre slide. It holds for all four families, as stated on the next slide, and then lets us expand reasonable functions exactly as Fourier series expand in e^{inx} .

Why the expansions converge: completeness

Orthogonality finds the coefficients. Completeness says that the full family can approximate every function with a finite weighted squared norm.

For any of the four families, let ϕ_n denote its polynomials and use the interval I and weight w from the table. Then

$$c_n = \frac{\int_I f(x) \phi_n(x) w(x) dx}{\int_I \phi_n(x)^2 w(x) dx}, \quad S_N(x) = \sum_{n=0}^N c_n \phi_n(x).$$

The completeness result, used here without proof, is

$$\int_I |f(x) - S_N(x)|^2 w(x) dx \longrightarrow 0 \quad \text{whenever} \quad \int_I |f(x)|^2 w(x) dx < \infty.$$

This is *mean-square* convergence: the total weighted squared error tends to zero. It does not assert convergence at every individual point.

The common workflow. Choose the family from the differential equation and its interval; identify the weight; compute coefficients by orthogonality; use a finite sum as an approximation.